

TORCS AI Racing Setup Guide

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Prerequisites

Register, open and review IBM Granite credential - <https://ibm.biz/TORCSnewGraniteSW>

Register and Mark Complete your AI Race League Innovation Certificate - <https://ibm.biz/AIRaceLeagueCert>

Ensure you have:

- **Windows OS** (Mac OS requires use of Wine – see presentation for further instructions)
 - **Python installed** on your system
 - **TORCS package downloaded** from <https://ibm.biz/TORCSdownloadzip>
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Installation & Setup

Step 1: Extract TORCS

Unzip the downloaded `torcs.zip` file to `C:\torcs\`

After extraction, your folder structure should be:

```
C:\torcs
├── gym_torcs
└── torcs\
```

Step 2: Launch TORCS Simulator

1. Navigate to `C:\torcs\torcs\` and run `wtorcs.exe`
2. In the TORCS menu, select: **Race** → **Quick Race** → **Configure Race**

Step 3: Configure Your Race Track

Selection:

- Choose any track you prefer **Driver**

Selection:

- **Must select:** `scr_server 1` (this is your AI-controlled car)
- Optional: Add other drivers for competition Click

New Race to proceed

Step 4: Start the AI Driver

While TORCS is loading the race:

- 1. Navigate to `C:\torcs\gym_torcs\`
- 2. Run `torcs_jm_par.py`

 **The race should now start with your AI driver in control!**

Controlling Your AI Car

The `torcs_jm_par.py` script establishes a UDP connection with TORCS to exchange telemetry data and control commands in real-time.

Core Functionality

- **Receives** sensor data from the simulator (speed, track position, wheel spin, etc.)
 - **Sends** driving commands back to TORCS (steering, throttle, brake, gear changes)
- Driving Modes**

The script includes two driving modes:

Mode	Description
<code>drive_example()</code>	Basic driving logic for getting started
<code>drive_modular()</code>	Advanced logic with customizable parameters

Main Loop

The driver operates continuously:
receive sensors → compute actions → send commands → repeat until race completion

Tuning Your AI Driver

User-Configurable Parameters

Experiment with these settings in the 'User Configurable Parameters' section of `torcs_jm_par.py` :

Parameter	Description	Tuning Tips
<code>TARGET_SPEED</code>	Target speed the car accelerates toward	Increase for faster racing, decrease for safer cornering
<code>STEER_GAIN</code>	Controls how sharply the car turns into corners	Higher = sharper turns, lower = smoother turns
<code>CENTERING_GAIN</code>	Controls how aggressively the car returns to the center line	Adjust for track positioning preference

Parameter	Description	Tuning Tips
BRAKE_THRESHOLD	Determines when the car should brake before corners	Lower = earlier braking, higher = later braking
GEAR_SPEEDS	Defines the speed thresholds for gear shifts	Optimize for your car's power band
ENABLE_TRACTION_CONTROL	Prevents wheel spin during acceleration	Recommended to keep enabled for stability

Applying Changes

1. Edit the parameters in `torcs_jm_par.py`
2. Save the file
3. Restart from **Step 4** above to test your changes

Troubleshooting

- **AI car not moving?** Ensure `torcs_jm_par.py` is running before the race starts
- **Connection issues?** Verify `scr_server 1` is selected in the driver configuration
- **Performance problems?** Try adjusting `TARGET_SPEED` and `BRAKE_THRESHOLD` first